

Gayatri Pooja Pothula

Data Analyst

Location: Colorado, USA | **Mail:** gayatri.pothula99@gmail.com | **Ph.:** (720) 408-4099 | [LinkedIn](#)

PROFESSIONAL SUMMARY:

- Data Analyst with over 6 years of progressive experience in leveraging complex datasets to drive strategic decision-making and operational efficiencies across diverse sectors including Healthcare, Banking and Telecommunications.
- Proven expertise in end-to-end data lifecycle management, encompassing robust data collection, meticulous cleaning, complex integration and advanced modeling from disparate sources such as Epic, Cerner, Meditech EMR, X12 EDI claims, Facets, QNXT, Finacle and Call Detail Records (CDRs).
- Adept at engineering scalable data warehouse solutions utilizing Snowflake, Azure Synapse and AWS Redshift, with a strong command of Star/Snowflake schema design for optimized query performance.
- Expert in developing and automating efficient ETL workflows using SSIS, Talend, Informatica and Unix/Linux shell scripting.
- Proficient in Python (scikit-learn, Statsmodels, XGBoost) and R (dplyr, ggplot2) for intricate data manipulation, advanced statistical analysis (including regression, survival and time-series analysis) and building impactful predictive machine learning models for credit risk assessment, customer churn and readmission.
- Skilled in processing large-scale datasets with Apache Spark and managing data within Hadoop HDFS.
- Committed to maintaining exceptional data accuracy and integrity by implementing stringent PHI data validation frameworks and comprehensive data quality (DQ) checks.
- Proven ability to design and deploy dynamic, interactive dashboards with Tableau, Power BI and QlikView to visualize critical KPIs and facilitate data-driven insights for leadership and operational teams.
- Experienced in ensuring full compliance with industry standards and regulations, including HIPAA, RBI and Basel III, while collaborating effectively with cross-functional teams using JIRA and Confluence.

TECHNICAL SKILLS:

Data Analysis & BI Tools:

Tableau, Power BI, QlikView, Excel (Advanced Formulas, Pivot Tables, Power Query, VBA), SSRS, Crystal Reports

Databases & Querying:

SQL Server, Oracle, MySQL, PostgreSQL, Teradata, Snowflake, AWS Redshift, Azure Synapse, MongoDB, DynamoDB

ETL & Data Integration:

SSIS, Talend, Informatica, Hadoop HDFS, Apache Spark, Unix/Linux Shell Scripting

Programming & Statistical Analysis:

Python (Pandas, NumPy, Scikit-learn, Statsmodels), R (dplyr, ggplot2), XGBoost, Logistic Regression, Time-Series & Survival Analysis

Healthcare & Financial Data Systems:

Epic, Cerner, Meditech EMR, Facets, QNXT, X12 EDI (837/835), ICD-10, CPT, HIPAA, Basel III, RBI Compliance

Data Warehousing & Modeling:

Star/Snowflake Schema Design, Data Quality (DQ) Frameworks, Data Governance, Data Lineage Documentation

Machine Learning & Predictive Modeling:

Risk Stratification, Churn Prediction, Credit Risk Assessment, Customer Segmentation

Cloud & Big Data Platforms:

AWS, Azure, Databricks, Apache Spark Streaming

Collaboration & Agile Tools:

Jira, Confluence, Git, MS Visio, MS Project

Other Skills:

Ggplot, Jupyter Notebook, decision tree, linear regression, Data Storytelling, Data Wrangling, Regression analysis, A/B testing, Data Modeling, Data Cleaning, Data Transformation, Data Visualization, Data warehousing

Operating Systems:

Windows, Linux, Mac iOS

PROFESSIONAL EXPERIENCE:

Banner Health – AZ | January 2023 – Present

Data Analyst

- Collected, cleaned and integrated structured and unstructured healthcare data from Epic, Cerner and Meditech EMR systems using SQL and Python, ensuring data completeness and standardization across multiple facilities.
- Integrated X12 EDI 837/835 claims data into centralized reporting pipelines, maintaining HIPAA compliance and improving reimbursement cycle visibility by 28%.
- Engineered Snowflake-based data warehouse models (Star/Snowflake schema) to support advanced reporting and optimized query performance for population health analytics.
- Developed automated ETL workflows using SSIS and Talend to ingest data from claims (Facets, QNXT) and EHR repositories, reducing manual processing time by 40%.
- Achieved 99.4% data accuracy by implementing robust PHI data validation frameworks and data quality (DQ) checks using Python, improving downstream reporting reliability.
- Designed and deployed interactive Tableau dashboards for hospital leadership to track KPIs such as ALOS, readmission rates and patient throughput, enabling data-driven decisions.

- Built risk stratification and readmission prediction models using scikit-learn and logistic regression techniques, improving chronic disease management and reducing readmission risk by 31%.
- Conducted regression, survival and time-series analyses using Statsmodels and R to forecast patient outcomes and optimize resource planning for critical departments.
- Led root cause analysis of denied claims using SQL-based analytics, uncovering ICD-10/CPT coding gaps and driving a 17% improvement in claim acceptance rates.
- Maintained data lineage and governance documentation to meet HIPAA standards, while collaborating with clinicians, billing and operations teams to align KPIs and reporting deliverables.

HCLTech – India | Sept 2020 – Dec 2021

Data Analyst

- Collected and preprocessed large-scale customer credit, transaction and demographic datasets from Finacle and internal banking databases, ensuring high-quality inputs for analytics initiatives.
- Developed and implemented rigorous data quality checks, adhering to industry data governance standards to maintain data integrity across all banking analytics processes.
- Utilized SQL (Oracle) for complex data extraction and transformation from core banking systems, enabling efficient data flow and preparation for downstream analysis.
- Built and managed a data warehouse using Azure Synapse to support scalable storage, fast retrieval and unified management of credit risk and financial data.
- Engineered features from structured and unstructured financial data to enhance predictive model performance, tailored to banking sector challenges.
- Designed and deployed machine learning models using Python (Scikit-learn) for credit risk assessment, focusing on the prediction of loan default probabilities.
- Conducted advanced statistical analysis in R (dplyr, ggplot2) to uncover key risk drivers, portfolio trends and actionable insights aligned with regulatory and business objectives.
- Developed interactive dashboards in Power BI for visualizing credit risk metrics, enabling real-time decision support for risk management and compliance teams.
- Collaborated with risk management teams to define, validate and adjust risk appetite and tolerance levels, ensuring model outputs are operationally relevant and regulatory compliant.
- Ensured full compliance with RBI and Basel III regulatory requirements through continuous monitoring, calibration and documentation of data models, analytical processes and reports.

Capgemini – India | June 2018 - August 2020

Data Analyst

- Analyzed Call Detail Records (CDRs) using SQL on Teradata to uncover usage patterns and pinpoint network performance issues, enabling proactive optimization of service quality.
- Developed predictive models for customer churn using Scikit-Learn and XGBoost, achieving 85% accuracy and reducing churn by 15% through targeted retention strategies.
- Optimized network uptime by 12% by identifying KPI bottlenecks with Ericsson ENIQ, facilitating timely interventions to enhance service reliability.
- Built ETL pipelines using Informatica to integrate OSS/BSS data into Hadoop HDFS, streamlining data processing and reducing latency by 20%.
- Performed customer segmentation with clustering algorithms to enable personalized marketing campaigns, boosting ARPU by 8% through tailored offers.
- Designed interactive dashboards in QlikView to visualize network KPIs and KQIs, empowering stakeholders with actionable insights for decision-making.
- Conducted time series forecasting using R to predict network traffic, optimizing resource allocation and improving capacity planning.
- Processed large-scale CDR datasets with Apache Spark for real-time analytics, enhancing the ability to monitor network performance dynamically.
- Automated data extraction and reporting using Unix/Linux shell scripting, cutting manual effort by 30% and improving operational efficiency.
- Ensured data quality by implementing validation frameworks for CDR and billing datasets, maintaining consistency and reliability for downstream analytics.
- Collaborated with cross-functional teams via JIRA and Confluence to align analytics initiatives with business goals, ensuring seamless project execution.
- Leveraged AWS Redshift for scalable querying of telecom datasets, enabling faster insights and supporting data-driven strategies for network and customer management.

EDUCATION:

Master of Business Analytics - University of Colorado, Denver, CO, USA

Bachelor of Business Administration - St. Joseph's Degree College, Kurnool, AP, India